GEOGRAPHY

PHYSICAL GEOGRAPHY

CLIMATOLOGY

MODULE – 23 | **EL-NINO AND ENSO EVENTS**

EL-NINO

WHAT IS IT ACTUALLY?

1. El Niño is the name given to the occasional development of warm ocean surface waters along the coast of Ecuador and Peru.
2. When this warming occurs the usual upwelling of cold, nutrient rich deep ocean water is significantly reduced.

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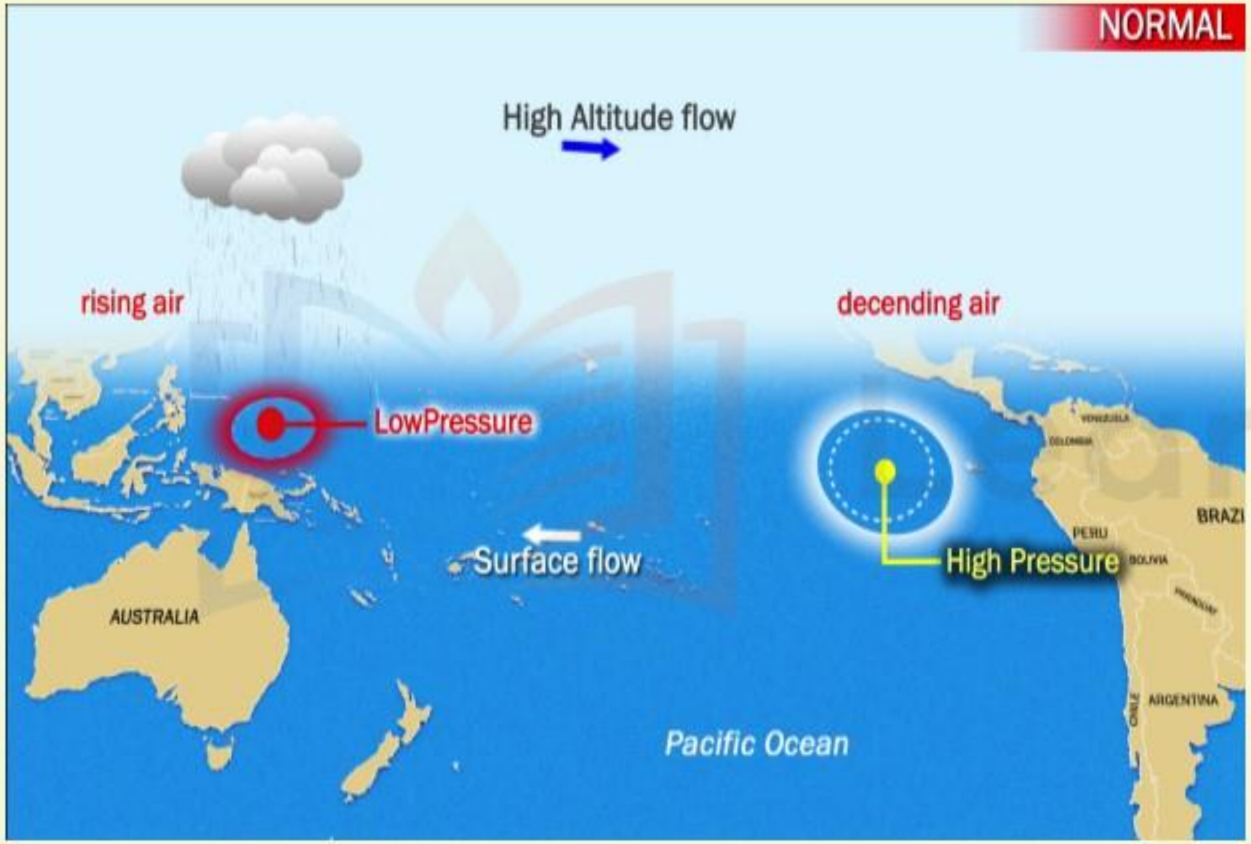
EL-NINO

WHAT HAPPENS DURING NORMAL CONDITIONS?

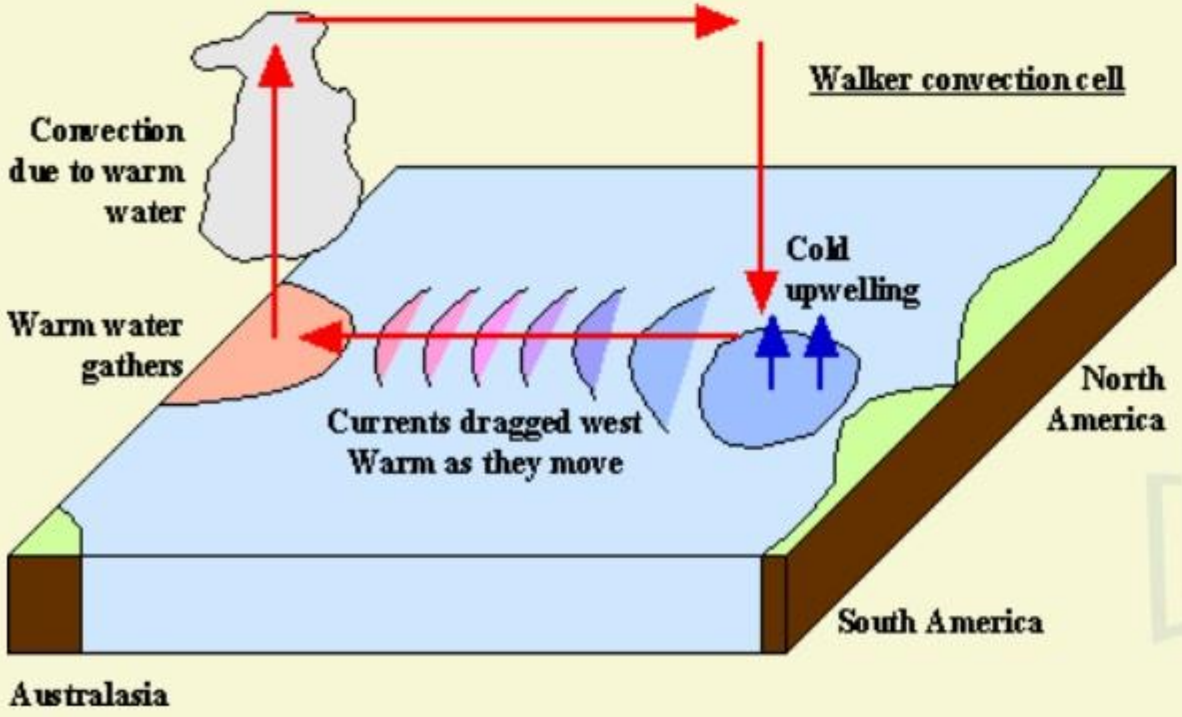


1. In a normal year, a surface low pressure develops in the region of northern Australia and Indonesia and a high pressure system over the coast of Peru.
2. As a result, the trade winds over the Pacific Ocean move strongly from east to west.
3. The easterly flow of the trade winds carries warm surface waters westward, bringing convective storms (thunderstorms) to Indonesia and coastal Australia.
4. Along the coast of Peru, cold nutrient rich water wells up to the surface to replace the warm water that is pulled to the west.


VENEZUELA
COLOMBIA
BRAZI
PERU
BOLIVIA
PARAGUAY
AUSTRALIA
CHILE
ARGENTINA
Pacific Ocean

EL-NINO

WHAT DO YOU MEAN BY WALKER CIRCULATION



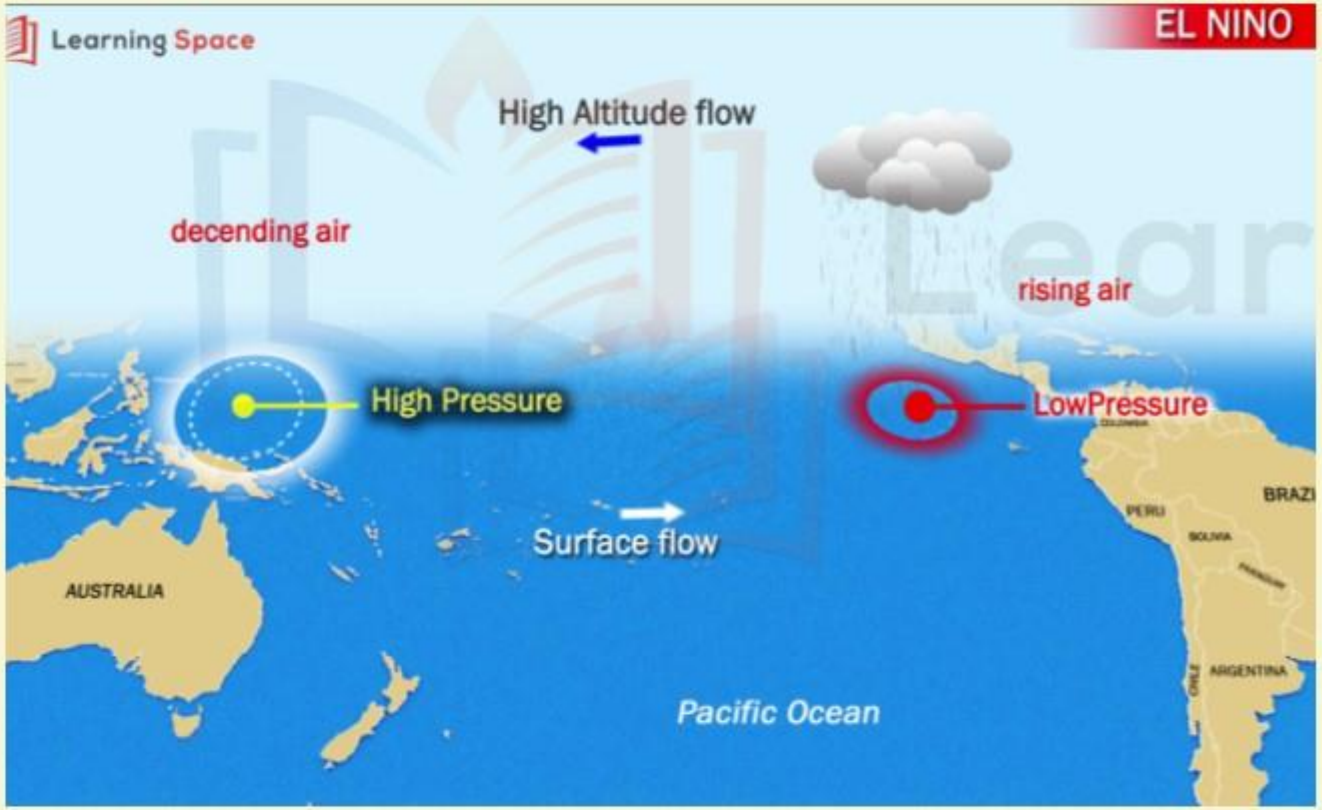
1. The Walker circulation (walker cell) is caused by the pressure gradient force that results from a high pressure system over the eastern Pacific ocean, and a low pressure system over Indonesia.

2. The Walker cell is indirectly related to upwelling off the coasts of Peru and Ecuador.

3. This brings nutrient-rich cold water to the surface, increasing fishing stocks.

EL-NINO

WHAT HAPPENS DURING AN EL-NINO YEAR?



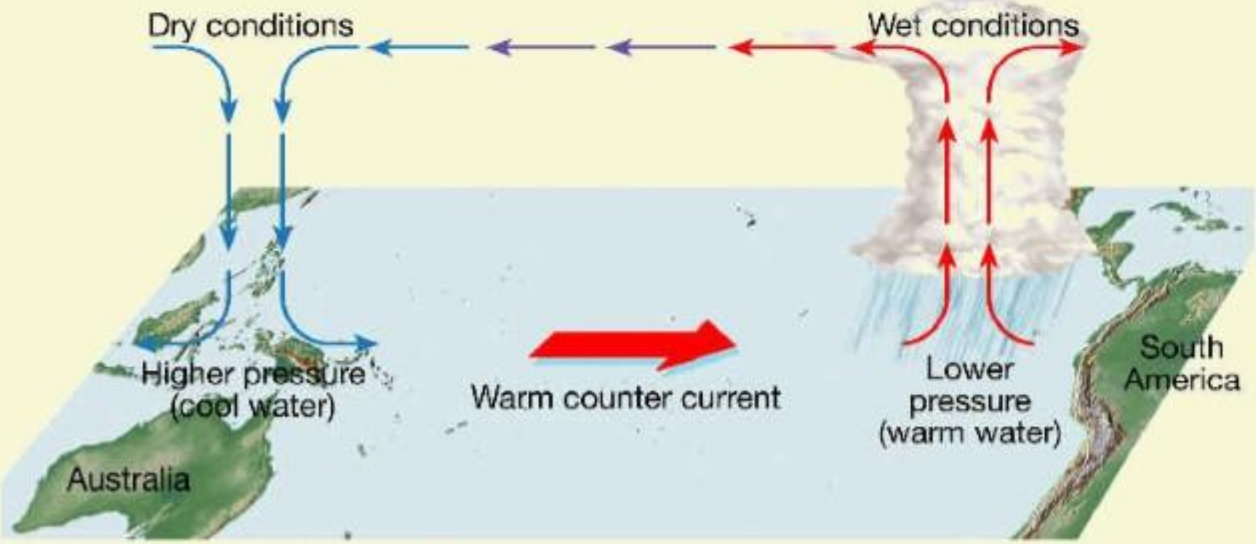
1. In an El Niño year, air pressure drops over large areas of the central Pacific and along the coast of South America.
2. The normal low pressure system is replaced by a weak high in the western Pacific (the southern oscillation).
3. This change in pressure pattern causes the trade winds to be reduced in intensity ie Weak Walker Cell.
4. Sometimes Walker Cell might even get reversed.

Learning Space

EL NINO

EL-NINO

WHAT HAPPENS DURING AN EL-NINO YEAR?



1 This reduction allows the equatorial counter current (current along doldrums) to accumulate warm ocean water along the coastlines of Peru and Ecuador.

2 This accumulation of warm water in the eastern part of Pacific Ocean which cuts off the upwelling of cold deep ocean water along the coast of Peru.

Climatically, the development of an El Niño brings drought to the western Pacific, rains to the equatorial coast of South America, and convective storms and hurricanes to the central Pacific.

EL-NINO

WHAT ARE THE EFFECTS OF EL-NINO?

1. The warmer waters has a devastating effect on marine life existing off the coast of Peru and Ecuador.
2. Fish catches off the coast of South America were lower than in the normal year (Because there is no upwelling).
3. Severe droughts occur in Australia, Indonesia, India and southern Africa.

El Nino → ✗ Upwelling → ✗ Phytoplankton → ✗ anchovies → ✗ guano

EL-NINO

WHAT ARE THE EFFECTS OF EL-NINO?

1. El Nino and Indian monsoon are inversely related.
2. The most prominent droughts in India have been El Nino droughts.
3. However, not all El Nino years led to a drought in India. For instance, 1997/98 was a strong El Nino year but there was no drought (Because of IOD).
4. El Nino directly impacts India's agrarian economy as it tends to lower the production of summer crops such as rice, sugarcane, cotton and oilseeds.

The ultimate impact is seen in the form of high inflation and low gross domestic product growth as agriculture contributes around 17 per cent to the Indian economy.

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EL-NINO

EL NINO SOUTHERN OSCILLATION [ENSO]

1. The formation of an El Niño is linked with Pacific Ocean circulation pattern known as the southern oscillation [circulation of atmospheric pressure]

2. Southern Oscillation is a coherent inter-annual fluctuation of atmospheric pressure over the tropical Indo-Pacific region.

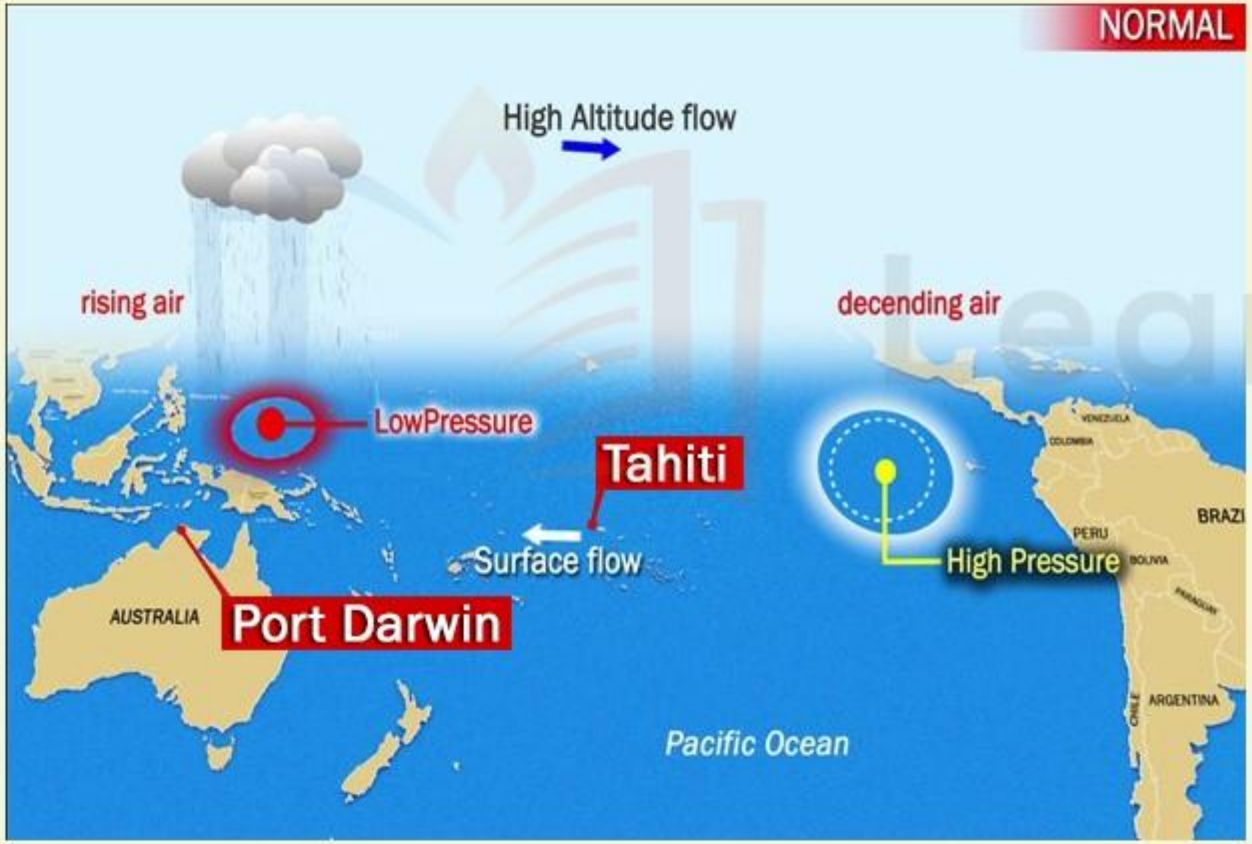
3. El Nino and Southern Oscillation coincide most of the times hence their combination is called ENSO– El Nino Southern Oscillation.

4. The pattern of low and high pressures gives rise to vertical circulation with surface flow and atmospheric flow leading to Southern Oscillations and due to its close association with El-Nino development, it is combinedly called as ENSO event.

5. Thus, ENSO can be said to be a best example of atmospheric-oceanic telecommunication.

EL-NINO

SOUTHERN OSCILLATION INDEX AND INDIAN MONSOONS



1. Southern Oscillation Index (SOI) is used to measure the intensity of the Southern Oscillation.
2. This is the difference in pressure between Tahiti in French Polynesia representing the Central Pacific Ocean and Port Darwin in northern Australia representing the Western Pacific Ocean.
3. The positive and negative values of the SOI i.e. Tahiti minus the Port Darwin pressure are pointers towards good or bad rainfall in India.

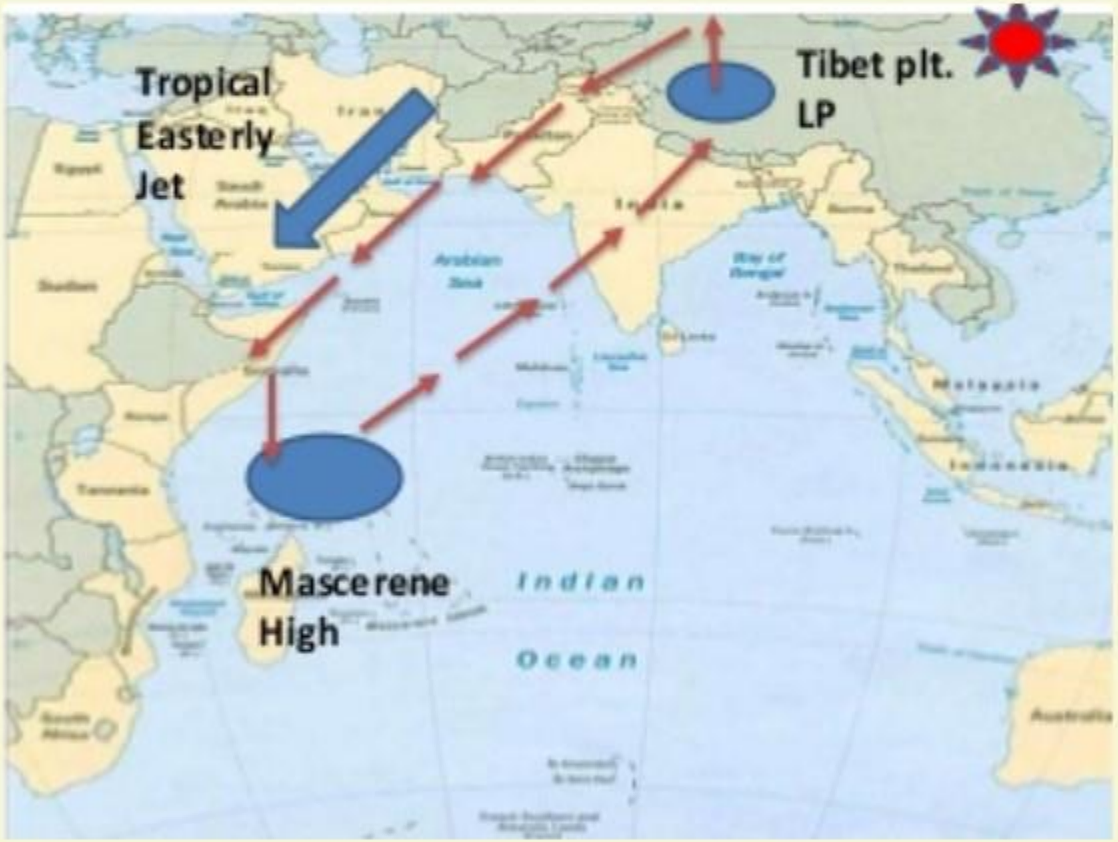
EL-NINO

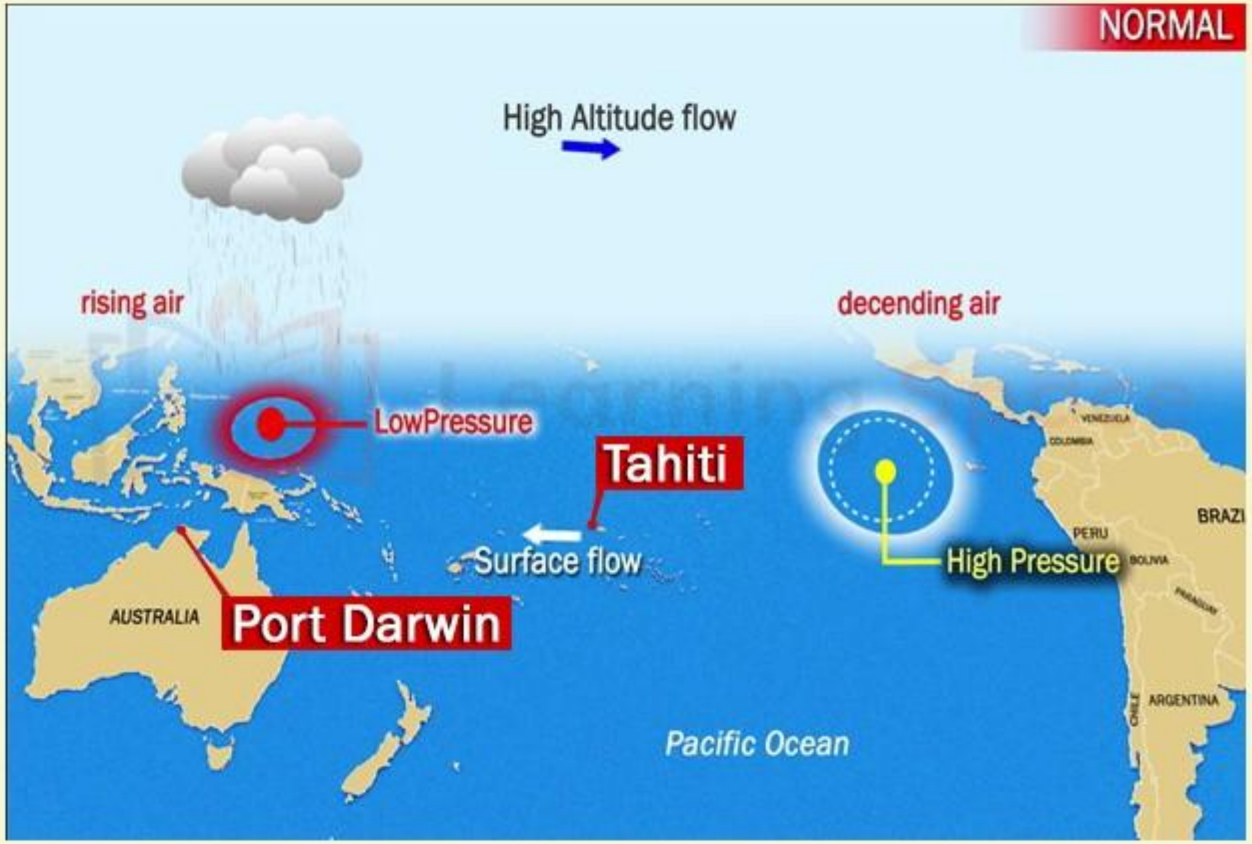
SOUTHERN OSCILLATION INDEX AND INDIAN MONSOONS

POSITIVE SOI
Tahiti pressure is greater than Port Darwin
Drought conditions in East Pacific and good rainfall in West Pacific (Northern Australia and Indonesia)
Good for Indian Monsoons

EL-NINO

SOUTHERN OSCILLATION INDEX AND INDIAN MONSOONS





CONCEPTS IN FUTURE MODULES IN INDIAN GEOGRAPHY RELATED TO ELNINO

1. INDIAN OCEAN DIPOLE
2. ENSO THEORY ON ORIGIN OF INDIAN MONSOONS
3. EFFECT OF TROPICAL EASTERLY JET/SUBTROPICAL WESTERLY JETSTREAM ON INDIAN MONSOONS

(IN COMPLETE ORIENTATION WITH NCERT BOOK INDIAN PHYSICAL GEOGRAPHY AND FUNDAMENTALS OF PHYSICAL GEOGRAPHY)

UPSC PRELIMS 2017 QUESTION

With reference to 'Indian Ocean Dipole (IOD)' sometimes mentioned in the news while forecasting Indian monsoon, which of the following statements is/are correct?

1. IOD phenomenon is characterized by a difference in sea surface temperature between tropical Western Indian Ocean and tropical Eastern Pacific Ocean.
2. An IOD phenomenon can influence an El Nino's impact on the monsoon.

Which of the above are correct?

a) 1 only
b) 2 only
c) Both 1 & 2 only
d) Neither 1 nor 2

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EL-NINO

UPSC 2014 GS-1 QUESTION

Most of the unusual climate happenings are explained as an outcome of the El-Nino effect. Do you agree? (10 Marks) (150 words)

HINTS

1. Define El-Nino.
2. Give a brief description of how it causes a HP in WPP and LP near S.America.
3. Causes excessive rainfall off Equador affecting phytoplankton growth- livelihood of people; food chain of Guano-Anchovies.
4. Recent droughts in South Africa, Zimbabwe; outbreak of diseases like hepatitis and cholera in Namibia.
5. Forest fires in Asia, South-East Asia.
6. Cyclonic storms driven away from the usual rainforest areas of Indonesia, Phillippines towards the drier Eastern Pacific.
7. Droughts and below-par rainfall in India and Australia.
8. Coral bleaching off the coast of Australia, New Zealand.
10. Conclusion: not just El-Nino but Human-driven extreme events also contribute.

EL-NINO

MOCK QUESTION

In view of the El-Nino, what immediate and long-term steps can be taken to mitigate its adverse effects?

HINTS

1. Think of all the problems that could arise because of El-Nino and come up with their possible solutions.
2. Food crisis; agricultural and poultry crisis- loss of food and livelihood
3. Power crisis because of weak monsoons
4. Economic crisis- RBI goes for stricter monetary policy because of food inflation; double blow to the country.
5. Water crisis- droughts due to El-Nino.
6. Key words to ponder- APMC Act, PDS, Farm Insurance Schemes, Watershed Management, drought-resistant seeds, financial inclusion.

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